

CODE SPORT



In this month's column, let's begin a discussion on neural networks.

Last month, we discussed computer science interview questions. Subsequently, a few of our readers have requested me to cover neural networks in this column. This topic has attracted a lot of interest and attention lately. It is a vast subject and hence it is extremely difficult to do justice to it in a short article. So what I plan to do over the next few months is to provide you a bird's eye view of neural networks, while pointing out additional resources where you can find more information.

While neural networks can be applied to different applications such as image recognition, speech processing, etc, we will discuss neural networks with a focus on natural language processing. Let's start off with a quick overview of the application of neural networks in natural language processing.

As the name indicates, the term neural network was indeed inspired by biological computational units in the brain known as neurons. While there is some exploratory research that is currently taking place on how the biological processes of brain computation can be simulated using neural networks, there is no connection between the biological neurons and the computer science neurons in any of the currently available neural network architectures used in natural language processing. Hence the neural networks of computer science were referred to as artificial neural networks, though of late, the term 'artificial' is no longer explicitly mentioned.

A single neuron is a computational unit with scalar inputs and outputs. There is a weight associated with each input to the neuron. The neuron multiplies each input by the associated weight and then sums up all the results. The result is then passed through a non-linear function whose result is the output of the neuron. Figure 1 shows a computation performed by a single neuron. The

non-linear function is also known as the activation function of the neuron. The weights associated with the inputs are also known as synaptic weights in connection with the synapse in the human brain, over which electrical/chemical impulses pass from one neuron to another.

A number of neurons can be connected to each other, forming a neural network. The output of one neuron is connected as input to one or more other neurons forming a connected network. These connections can be organised in the form of layers, with outputs of neurons in one layer providing inputs to neurons in the next layer. The output of a neuron in one layer can be connected to one or all of the neurons in the next layer. The network in which each neuron's output from one layer is connected to all other neurons in the next layer is known as a fully connected neural network. Neural networks typically contain one input layer, one output layer and one or more intermediate layers. The intermediate layers are also known as hidden layers.

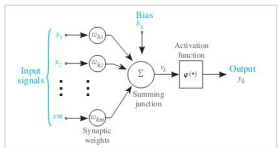


Figure 1: Computation performed by a single neuron

The simplest neural network is a network that contains one input layer, one intermediate layer/hidden layer and one output layer. An example of such a network is shown in Figure 2. Networks with